YouTube Music on Raspberry Pi: Setup Guide

A phone-controlled YouTube Music player for the Pi (ytm.py): search, playlists, downloads and local songs, played through mpv.

What you get

- A phone page at `http://<pi-ip>:8001` to search songs, albums and playlists, or open any public playlist by link.
- Queue, favourites, saved playlists and Auto Play after a reboot.
- Downloads as Opus into `~/YTMusic/DownloadedSongs`, plus playback of local files.
- System volume control (same method as radio.py) with a peak limiter to avoid distortion.
- No Google login or cookies needed.

Requirements

- A Raspberry Pi (built and tested on a Pi 5) running Raspberry Pi OS Bookworm or newer, desktop version, which uses PipeWire for audio.
- Internet access and working audio output (speakers, DAC, headphones or HDMI).
- SSH access, or a keyboard and screen, and the `ytm.py` file.

1. Update the system

```
sudo apt update && sudo apt full-upgrade -y
```

2. Install packages

```
sudo apt install -y mpv ffmpeg python3-pip nodejs
```

- **mpv** plays the audio.
- **ffmpeg** lets downloads become tagged `.opus` files. Without it they are saved as `.webm` (Opus inside), which still plays.
- **nodejs** is the JavaScript runtime yt-dlp may ask for. If yt-dlp still complains about a JavaScript runtime, install Deno instead.

3. Install the Python packages

```
pip install -U flask ytmusicapi yt-dlp --break-system-packages
echo 'export PATH=$HOME/.local/bin:$PATH' >> ~/.bashrc && source ~/.bashrc
yt-dlp --version
```

The last command should print a version number. If it says command not found, open a new terminal and try again.

4. Copy the app to the Pi

```
mkdir -p ~/YTMusic
```

From your PC, send the file (change `pi` and the address to match your Pi):

```
scp ytm.py pi@<pi-ip>:~/YTMusic/
```

Find the Pi's address by running `hostname -I` on the Pi.

5. Check audio and yt-dlp

```
wpctl status
wpctl set-volume -l 1.0 @DEFAULT_AUDIO_SINK@ 75%
mpv --no-video "https://www.youtube.com/watch?v=dQw4w9WgXcQ"
```

If you hear the track, audio and yt-dlp both work. Press `q` to quit mpv. Keep the system volume around 70 to 80% so there is headroom and no distortion.

6. First run

```
cd ~/YTMusic && python3 ytm.py
```

On your phone (same Wi-Fi), open `http://<pi-ip>:8001`. Press `Ctrl+C` on the Pi to stop it.

7. Start automatically on boot

```
(crontab -l 2>/dev/null | grep -v ytm.py; echo "@reboot sleep 40 && python3
/home/pi/YTMusic/ytm.py > /home/pi/YTMusic/ytm.log 2>&1") | crontab -
```

- Change `/home/pi` if your username is different.

- The 40-second delay gives the audio service time to start.
- The `grep -v` removes any older ytm.py line, so running this twice will not create duplicates.
- Check with `crontab -l`, then reboot once to confirm the page comes up and plays.
- To start it now without rebooting: `nohup python3 ~/YTMusic/ytm.py > ~/YTMusic/ytm.log 2>&1 &`

Optional settings

Set these in front of `python3` in the command or cron line, for example `YTM_NORMALIZE=1 python3 ...`.

Variable	Purpose
`MPV_AUDIO_DEVICE`	Force an audio device, for example `pipewire`. List names with `mpv --audio-device=help`.
`YTM_NORMALIZE=1`	Even out loudness between tracks.
`YTM_PORT`	Use a different port than 8001.
`YTM_DIR`	Use a different data folder than `~/YTMusic`.

Using the app

- **Search:** songs, videos, albums or playlists. Tap a result to play it, and the rest of the list follows. + adds to the queue, ♥ saves a favourite.
- **Playlists:** paste a public playlist or song link, then Play all, Shuffle, Save or ⬇ **All** to download everything.
- **Local:** songs in `DownloadedSongs`, with filter, Play, Shuffle and delete. Download progress shows at the top.
- **Favs:** your favourite tracks.
- **Queue:** jump to any track and switch Auto Play on or off.
- **Player bar:** previous, play/pause, next, seek, and volume − / + with mute. Volume controls the Pi's system volume.
- ⬇ on any track downloads the best available audio, one song at a time.

Files and folders

Path	What it is
`~/YTMusic/ytm.py`	The app
`~/YTMusic/DownloadedSongs/`	Downloaded and local songs. You can copy your own music here too.
`~/YTMusic/favourites.json`	Favourite tracks
`~/YTMusic/saved_playlists.json`	Saved playlists
`~/YTMusic/state.json`	Last queue and Auto Play setting
`~/YTMusic/mpv.log`	mpv messages, useful when tracks skip
`~/YTMusic/ytm.log`	App messages

Maintenance

```
# update the tools (about once a week)
pip install -U yt-dlp ytmusicapi --break-system-packages

# restart the app
pkill -f ytm.py
nohup python3 ~/YTMusic/ytm.py > ~/YTMusic/ytm.log 2>&1 &

# look at recent mpv messages
tail -40 ~/YTMusic/mpv.log
```

Keeping yt-dlp and ytmusicapi current fixes most breakage, since YouTube changes often.

Troubleshooting

Problem	Likely cause	Fix
Tracks skip instantly	No audio output. `mpv.log` shows `Host is down`.	Make sure the audio service is running (`wpctl status`), set `MPV_AUDIO_DEVICE`, and raise the cron delay to 60 seconds.

Problem	Likely cause	Fix
"Sign in to confirm you're not a bot" or HTTP 429	YouTube is throttling your IP	Wait a few hours, update yt-dlp, try another connection such as a phone hotspot, and stay off VPNs.
Search shows an error	ytmusicapi is outdated	Update it with pip (see Maintenance).
Warning about a JavaScript runtime	No runtime installed	`sudo apt install nodejs`, or install Deno.
Page does not load	App not running, or different network	Run `pgrep -af ytm.py`, check `ytm.log`, and use the same Wi-Fi as the Pi.
A download fails	See the message on the Local tab	Update yt-dlp and check free space with `df -h`.
Downloads are `.webm`	ffmpeg is not installed	`sudo apt install ffmpeg`. Only new downloads change.

Good practice

- Run without cookies. They are only worth adding if you get persistent blocks, and then only from a throwaway Google account, never your main one. Exported cookies also expire quickly and can cause "page needs to be reloaded" errors.
- Keep the load light: one stream at a time, no bulk downloading, and tools kept up to date.
- yt-dlp and ytmusicapi are unofficial tools. Use this for personal listening at home.
- The page has no password. Use it on your home network only and do not expose port 8001 to the internet.